



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

Faculty of
Computing

SEMESTER 1/20252026

ARTIFICIAL INTELLIGENCE (SECJ3553)

SECTION 03

GROUP 2

ASSIGNMENT 2:

EcoRoute

THEME:

SMART CITY

LECTURER:

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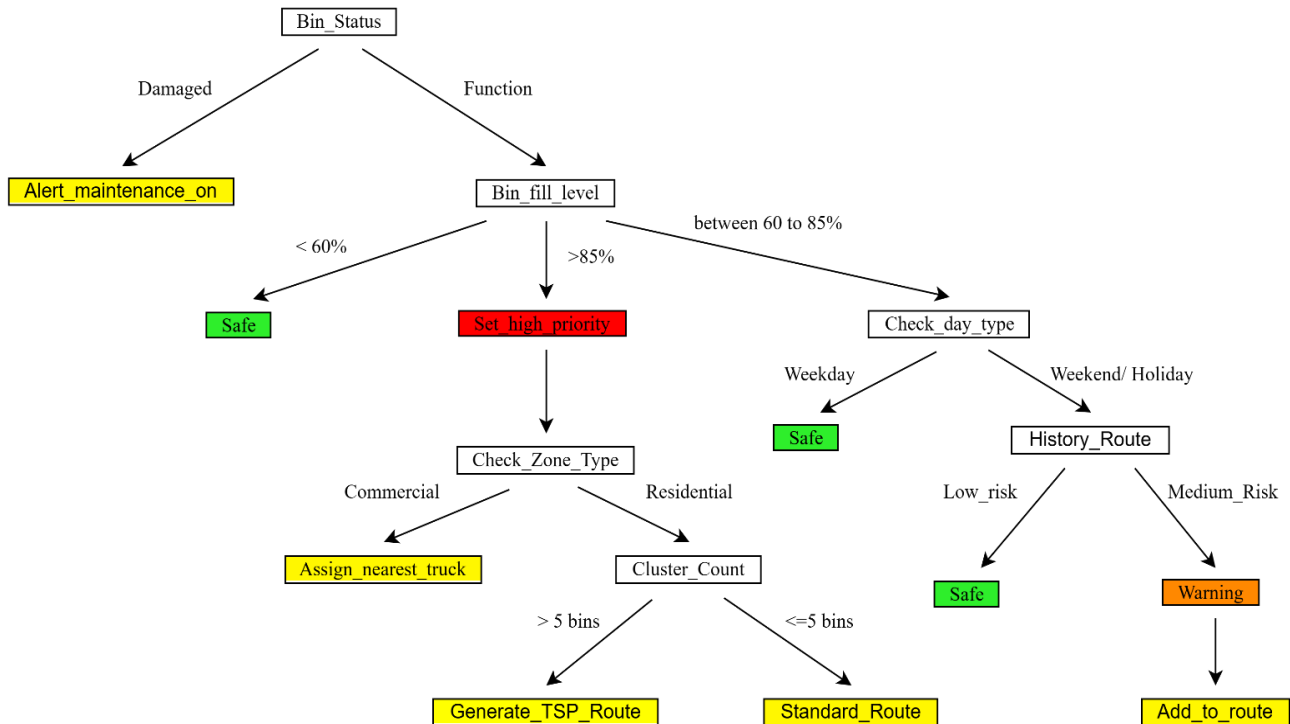
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1.0 Problem Formulation by State Space Graph

1.1 Bin Management



1.2 Explanation States and Actions

As proposed in our project proposal document and Assignment 1, EcoRoute is a waste management system based on AI that dynamically observes the condition of the bins, estimates the possibility of bin overflow, and provides the best routes to be followed by the trucks. The state space of the EcoRoute system can be described through a number of critical states. The transition from the states can be described through actions. The combination of states and actions guarantees efficient waste collection.

States	Bin_Status - Damaged: The bin cannot be serviced and requires maintenance before collection. This prevents wasted trips and ensures resources are not allocated to unusable bins. - Function: The bin is operational and can be monitored for fill levels, enabling the system to make real-time decisions. Bin_Fill_Level < 60%: Safe, no immediate collection required.
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	• 60 – 85%: Medium risk, monitored closely to prevent overflow. • > 85%: High priority, required immediate collection. This state directly implements the KR1 rule from Assignment 1 which is $\forall x((FillLevel(x) > 0.85) \wedge \neg bin_status(x, "Damaged")) \rightarrow AddRoute(x, HighPriority)$ 3. Day_Type • Weekday: Normal collection rules apply. • Weekend / Holiday: Higher risk of overflow due to increased waste generation. Predictive scheduling is triggered, lowering thresholds to act earlier. 4. Zone_Type • Commercial: Bins fill faster, lower threshold (70%) applied. • Residential: Bins fill slower, higher threshold (85%) applied. This ensures fairness and efficiency across different urban zones. 5. History_Route • Low Risk: Bin rarely overflows, standard scheduling applies. • Medium / High Risk: Bin frequently overflows, predictive scheduling required. This state uses historical data to anticipate problems before they occur. 6. Cluster_Count • ≤ 5 bins: Standard route assignment. • > 5 bins: Route optimization using the Traveling Salesperson Problem (TSP) algorithm. This ensures trucks take the shortest possible path when servicing multiple bins.
Action	1. Alert_Maintenance_On: Triggered when a bin is damaged. The system alerts maintenance teams to repair the bin before collection, preventing wasted fuel and time. 2. Set_High_Priority: Triggered when a bin's fill level exceeds 85%. The bin is marked for immediate collection, ensuring hygiene and preventing overflow. 3. Add_To_Route: Adds bins that meet threshold conditions to the truck's collection route, ensuring only necessary trips are made. 4. Assign_Nearest_Truck: Allocates the closest available truck to a bin, reducing travel time, fuel consumption and driver fatigue. 5. Generate_TSP_Route: When more than 5 high-priority bins are detected in a zone, the system applies the TSP algorithm to generate the most efficient route. This reduces distance traveled and operational costs. 6. Standard_Route: Used when fewer than 5 bins require collection, following normal scheduling.

1.3 Problem Formulation and Explanation

Critical	Details	Explanation
Initial state	Bin_Status = Function Bin_Fill_Level < 60% Day_Type = Weekday Zone_Type = Residential	The system starts with a bin that is operational, has a low fill level, and is in a standard monitoring state
Action	Alert_Maintenance_On Set_High_Priority Add_To_Route Assign_Nearest_Truck Generate_TSP_Route Standard_Route	These are the specific operations the system can perform to transition between states, such as flagging a bin for repair, changing its collection priority, or optimizing the truck's path.
Goal	Bin_Fill_Level < 60% Bin_Status = Function	The desired outcome is that all bins are emptied and remain functional, preventing overflow and ensuring system efficiency.
Path Cost	Distance traveled by trucks Fuel consumption Time consumption	Longer routes increase fuel usage and time consumption. Higher fuel usage increase overall operational cost. Longer time to complete bin collection lead to inefficiencies and delays.

Solution: sequence of actions leading from initial state to goal state

1. Monitor Bin_fill_level

- Ensure the bin is functional, and the fill level increases over time.

2. Set_high_priority

- When Bin_fill_level exceeds 85%, the system marked the bin as high priority.

3. Assign_nearest_truck

- The system assigns the bin to the nearest truck that is available to reduce travel costs.

4. Generate_TSP_Route

- Use TSP optimization when the high-priority bin in the zone is more than 5.

5. Truck Collects Waste

- Truck follows the route generated and empties the bin.

6. Update Bin Status

- After collection, Bin_Fill_Level is reset to < 60% and Bin_Status remains function.

Assignment 2 Peer Review Form

Evaluator: Choh Jing Yi

Name	Rating (1-4)	Notes/Comments (if any)
Brendan Chia Yan Fei	4	
Anis Safiyya Binti Janai	4	
Lim Yu Han	4	

Evaluator: Brendan Chia Yan Fei

Name	Rating (1-4)	Notes/Comments (if any)
Choh Jing Yi	4	
Anis Safiyya Binti Janai	4	
Lim Yu Han	4	

Evaluator: Anis Safiyya Binti Janai

Name	Rating (1-4)	Notes/Comments (if any)
Choh Jing Yi	4	
Brendan Chia Yan Fei	4	
Lim Yu Han	4	

Evaluator: Lim Yu Han

Name	Rating (1-4)	Notes/Comments (if any)
Choh Jing Yi	4	
Brendan Chia Yan Fei	4	
Anis Safiyya Binti Janai	4	